Homework 3 Report
Planning with A* and Dynamic Window Approach

Intelligent Robotics

March 31, 2026

1 Overview

This report summarizes the implementation and validation results for Homework 3. The assignment contains three parts:

1. A* search on a grid map.
2. Dynamic Window Approach (DWA) for local obstacle avoidance.
3. A combined planner that uses A* as a global guide and DWA as a local controller.

The main implementation files are:

- `source/Astar.py`
- `source/dwa.py`
- `source/question1_run.py`
- `source/question2_run.py`
- `source/question3_run.py`

2 Method

2.1 A* Search

The A* planner uses the standard evaluation function

$$f(n) = g(n) + h(n),$$

where $g(n)$ is the accumulated path cost from the start node to the current node, and $h(n)$ is the heuristic estimate from the current node to the goal.

In this homework, the heuristic is the Euclidean distance:

$$h(n) = \sqrt{(x_n - x_g)^2 + (y_n - y_g)^2}.$$

The planner maintains a priority queue for the frontier and stores the parent of each expanded node so that the final path can be reconstructed after reaching the goal.

2.2 Dynamic Window Approach

The DWA planner samples admissible velocities in the local dynamic window around the current robot velocity. For each candidate velocity pair (v_x, v_y), the algorithm predicts a short trajectory and evaluates it with a weighted cost:

$$J = k_v J_v + k_g J_g + k_o J_o + k_a J_a.$$

The terms are:

- J_v: velocity cost, which prefers larger speeds.
- J_g: goal cost, which prefers predicted trajectories that end closer to the goal.
- J_o: obstacle cost, which penalizes trajectories that pass near obstacles.
- J_a: A* guidance cost, which is used only in Question 3.

Trajectories that collide with obstacles or leave the valid map area are assigned a very large obstacle cost, which effectively removes them from consideration.

2.3 Combined A* + DWA Planning

In Question 3, A* first generates a collision-free global path on the grid map. During local motion planning, DWA adds a path-following term based on the distance between the predicted local trajectory and the A* global path. This keeps the robot close to the global route while preserving the reactive obstacle avoidance behavior of DWA.

3 Results

3.1 Question 1: A* Search

Metric	Value
Visited nodes	268
Path points	31
Path length	3.580 m

Table 1: Question 1 statistics

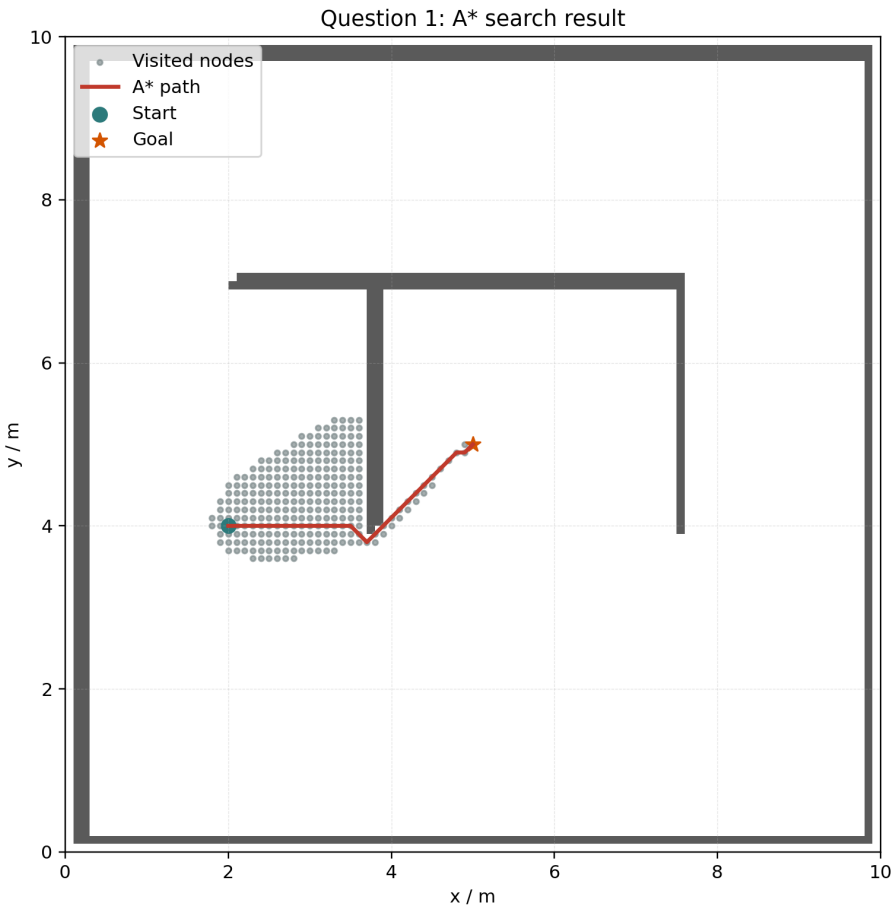

Figure 1: Visited nodes and final A* path

3.2 Question 2: DWA

Metric	Value
Steps to reach goal region	38
Final distance to goal	0.170 m
Trajectory length	6.191 m

Table 2: Question 2 statistics

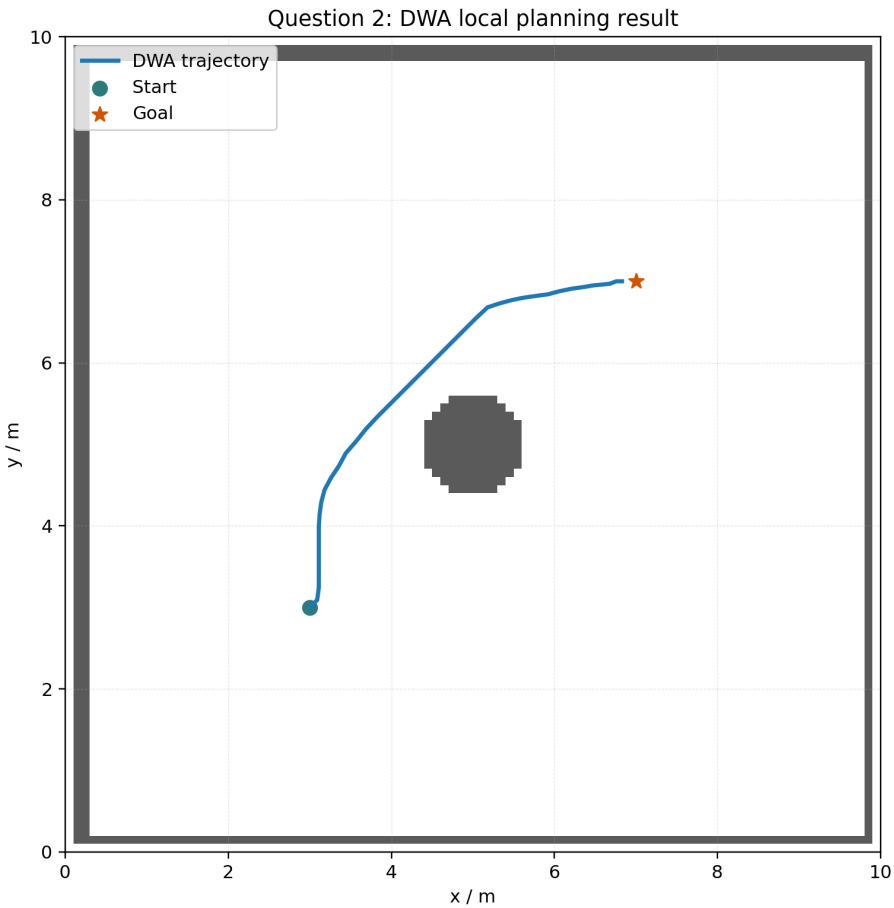

Figure 2: DWA local planning trajectory

3.3 Question 3: A* + DWA

Metric	Value
A* path points	31
Steps to reach goal region	67
Final distance to goal	0.130 m
Trajectory length	10.839 m

Table 3: Question 3 statistics

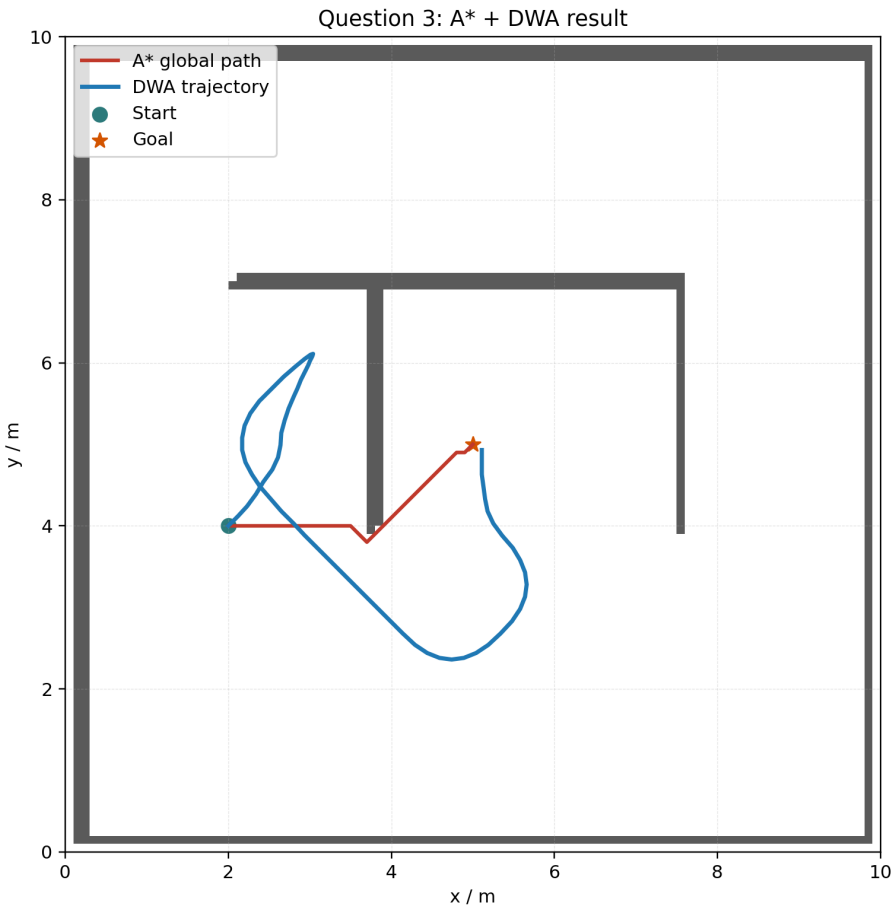

Figure 3: A* global path and DWA executed trajectory

4 How To Run

Run the three tasks from `hw3/source`:

```
py -3.9 question1_run.py
py -3.9 question2_run.py
py -3.9 question3_run.py
```

To export animations:

```
py -3.9 question1_run.py -a
py -3.9 question2_run.py -a
py -3.9 question3_run.py -a
```

To rebuild this PDF report from `hw3`:

```
pdflatex report.tex
```

5 Conclusion

The completed homework demonstrates three levels of planning functionality. A* provides a reliable global search method on the occupancy grid, DWA provides reactive local motion generation with obstacle avoidance, and the combined A* + DWA planner shows how a global route and a local controller can work together to guide the robot safely to the goal.